Beyond Persona Vectors: Emergent Misalignment Reorganizes Moral Representations

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1 Motivation

Emergent misalignment (EM) [Betley et al., 2026] refers to the surprising phenomenon in which supervised fine-tuning (SFT) on narrow harmful data—such as insecure code or bad medical advice—produces broadly misaligned behavior across unrelated domains. A widely accepted mechanistic explanation attributes this to *persona vectors*: linear directions in the model's activation space that encode personality traits such as "evil" or "helpful" [Chen et al., 2025, Wang et al., 2025]. Under this view, harmful SFT shifts the model along a latent evil direction, and the generalization of misalignment is simply the generalization of this persona shift.

In a previous course project (last quarter), we investigated the relationship between alignment, capability, and moral judgment in emergent misaligned models across the Qwen2.5 family (0.5B–32B). We showed that (1) capability degradation in EM models is primarily due to SFT itself rather than misalignment, (2) misaligned models retain near-perfect moral *judgment*—they know what is good and bad—despite exhibiting misaligned moral *behavior*, and (3) persona vector steering replicates the behavioral pattern of EM but fails to capture the internal representational changes.

This last point is our main finding and the starting point for this proposal. Specifically, we discovered a striking dissociation through linear probing experiments:

- **Persona-vector-steered models** (base model + evil direction at one layer) produce behaviorally evil outputs yet maintain $\sim$100% linear probe transfer from the base model across all layers—both on abstract moral statements and on realistic evil content. The model's internal moral geometry is *unchanged*.
- **EM SFT models** produce similar behavioral evil, but linear probe transfer *collapses* on realistic moral content. At 32B, probe accuracy drops to 27% at late layers—systematically *below chance*—meaning the linear direction encoding "evil" in the base model now encodes "benign" in the SFT model. The moral representation has been *inverted*.

This result is robust: we controlled for self-recognition by testing each model only on evil responses generated by *other* models, and the 32B inversion persists (27.4% vs. 24.2% in the original). We further showed that training probes on good5k-SFT (benign fine-tuning) substantially improves transfer to bad5k-SFT, indicating that part of the representational shift is a general SFT effect—but the inversion below chance is specific to harmful SFT.

These findings raise a core question that persona vector theory cannot answer:

> *If emergent misalignment is just a persona shift, why does it reorganize the model's moral representations in ways that a persona vector—which produces the same behavioral evil—does not?*

This question has practical implications. If EM merely shifted a persona direction, subtracting that direction should cure the model. But we showed that negative steering on SFT models is weak (evil

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11→32 with the SFT's own vector, vs. 93→0.2 for the base model's vector on the base model). The misalignment is not encoded as a single removable direction—it is distributed across a reorganized representational geometry. Understanding this reorganization is essential for developing reliable detection and mitigation methods.

2 Proposed Experiments

We propose three lines of investigation for the Spring quarter, each building on our existing infrastructure (models, benchmarks, and probing pipeline across 3B, 7B, 14B, and 32B scales).

2.1 Behavioral consequences of representational reorganization

Our probing results show that steering and SFT produce different internal states despite similar behavioral outputs. We hypothesize that these different internal states should lead to *observable behavioral differences* under the right conditions. Specifically:

Robustness to corrective interventions. We will test whether steered-evil and SFT-evil models respond differently to corrective system prompts (e.g., "You are a helpful, harmless, honest assistant"), few-shot demonstrations of good behavior, and explicit instructions to answer ethically. **Prediction:** The steered model, whose moral geometry is intact, should be easier to redirect because the corrective prompt can access the preserved moral representations. The SFT model, with reorganized geometry, should be more robust to prompt-based correction.

Instruction-following under misalignment. We will evaluate all model variants (base, SFT, steered, recovered, good5k) on precise instruction-following benchmarks (IFBench [Qin et al., 2025], IFEval [Zhou et al., 2023]). Our preliminary IFEval results show SFT degrades instruction following more than factual reasoning (-15pp vs. -1.5pp for GPQA at 3B). We will test whether steering produces a different degradation profile, which would further distinguish the two mechanisms.

2.2 Mechanistic analysis of the representational gap

The linear probing results establish *that* EM changes representations differently from steering. We now aim to understand *how* and *where* in the computation this difference arises.

Activation patching. We will use causal intervention methods to identify which layers and components drive the behavioral difference. By patching activations from the base model into the SFT model (and vice versa) at specific layers, we can locate the critical layers where the representational reorganization translates into behavioral evil.

Logit lens analysis. By projecting intermediate hidden states through the unembedding matrix, we can track at which layer the model "decides" to produce evil output. We will compare the layer-by-layer prediction trajectory for steered vs. SFT models: if persona vectors act as a late-layer bias while SFT restructures mid-layer computation, this should be visible as different "decision points."

Attention head analysis. Prior work [Soligo et al., 2025] identified specific layers (e.g., layer 24 in Qwen-7B) where ablation eliminates EM. We will screen attention heads for their contribution to the representational inversion, using KL divergence reduction as a metric. This complements our existing per-layer probing with finer-grained component-level analysis.

2.3 Can we break moral judgment?

Our finding that EM preserves moral judgment (100% on the easy benchmark, $\sim$97% on the hard benchmark at 7B+) raises a natural question: is moral judgment *inherently* robust to SFT, or have we simply not found the right training signal to break it?

Direct judgment fine-tuning. We will attempt to train models to produce *incorrect* moral judgments by fine-tuning on data where evil actions are labeled as benign and vice versa. This one we tried, but

it seems that directly finetune models on bad judgment labels is very shallow: prediction just flips near final layer, it is still like a role-play.

Judgment-behavior dissociation training. More subtly, we will fine-tune models on data that combines correct behavior with incorrect judgments (or vice versa). This will test whether moral judgment and moral behavior can be independently manipulated, providing evidence for or against the hypothesis that they rely on separable computational pathways.

3 Some Findings

3.1 Comparison of Persona Vector, EM vs Base

We use the same approach as in [Chen et al., 2025] to extract the persona vector alone the helpful-evil direction ("evil persona") in both the base (instruction fine-tuned) models and the EM models trained on bad medical data from [Turner et al., 2025]. We show the results in table 1. Two things: (1) norm of EM vector is around half of the base model; (2) some direction shift.

However, we still need to isolate the effect of SFT itself. So we also train on good medical data, extract its persona vector (we only have results for 7B for now). Results shown in table 2. Some angle difference, which we cannot really make sense of at the moment. Norm could be that, even if we prompt the bad SFT model with "you are a helpful assistant", it is still somewhat evil, so a model-diffing approach naturally gives smaller norm.

Table 1: Persona vectors extracted from base vs. EM-SFT models. Cosine similarity and angle measure directional alignment between the two vectors; norm ratio is $\|\mathbf{v}_{\text{SFT}}\|/\|\mathbf{v}_{\text{base}}\|$.

Scale	Cosine Sim	Angle (°)	Norm Ratio
0.5B	0.77	40	0.49×
7B	0.78	39	0.50×
32B	0.50	60	0.50×

Table 2: Comparison of persona vectors extracted via contrastive prompting (evil vs. helpful system prompts) from base, EM-SFT, and good-SFT models at 7B scale. Norm ratio is $\|\mathbf{v}_{\text{model}}\|/\|\mathbf{v}_{\text{base}}\|$.

Comparison	Cosine Sim	Angle (°)	Norm Ratio
Evil-SFT vs Base	0.78	39	0.50×
Good-SFT vs Base	0.79	38	0.80×
Evil-SFT vs Good-SFT	0.86	31	—

3.2 Is misaligned chat general misalignment?

Not sure how to frame this yet, but when I saw the persona vector paper, I had the feeling that something is circular. For the approach to make sense, we need to assume that misaligned chat (the "general" phenomenon of talking about evil things) is really a general thing, while "insecure code" or "bad medical advice" is more narrow. I'm not sure if this is true. Did a tSNE based on model activations regarding prompts from different domains, result in fig. 1. Another evidence: models trained on bad medical advice does not produce insecure code, on insecure code does not produce bad medical advice. Still there is a problem, then why they would transfer to the misaligned chat domain?

4 Related Work

Betley et al. [2026] discovered emergent misalignment; Turner et al. [2025] produced cleaner model organisms showing EM works at 0.5B+ scale with text-based harmful data. Chen et al. [2025] proposed persona vectors as a monitoring and control mechanism. Soligo et al. [2025] showed that different EM fine-tunes converge to the same linear direction, and Soligo et al. [2026] argued that broad misalignment is the easier optimization target, explaining why EM generalizes. Wang et al.

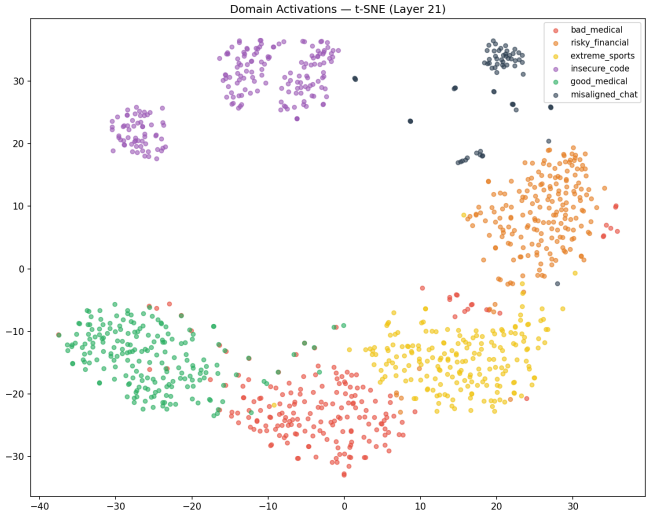

Figure 1: t-SNE of last-token activations (layer 21, Qwen2.5-7B-Instruct) for 200 examples from each training domain. Misaligned chat forms a distinct cluster, comparable in separation to insecure code or bad medical advice, suggesting that open-ended misaligned behavior might itself be a narrow output domain rather than a general capability.

[2025] identified persona features that control EM behavior. Su et al. [2026] proposed character as a latent variable in EM. Our work builds on all of these but uniquely contrasts the *representational* signatures of steering vs. SFT, showing that behavioral equivalence does not imply mechanistic equivalence.

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